

Dear Hiring Manager,

I am writing to apply for the position of **Specialist – System Engineer/System Development with ADAS (89373)**. I hold a Ph.D. in Mechanical Engineering from NIT Warangal and have professional experience as a Robotics Software Engineer, along with my current role as an Assistant Professor in Robotics and Automation. My background combines rigorous research with hands-on industry experience in autonomous mobile robotics, sensor fusion, localization, planning, control, ROS 2, and experimental validation.

I have extensive research and practical experience in the **development, integration, and experimental validation of autonomous mobile robot systems**. During my Ph.D., my research focused on coverage path planning for mobile robots operating in complex environments, including flat and uneven terrains. I developed improved variants of the **NSGA-II algorithm for multi-objective autonomous navigation**, addressing objectives such as travel distance, number of turns, energy efficiency, terrain-related cost, and obstacle avoidance. My research on multi-objective coverage path planning was published in *Robotica* by Cambridge University Press: <https://doi.org/10.1017/S0263574724000377>.

The developed approaches were evaluated through simulation and experimentally validated on a **differential-drive GoPiGo3 mobile robot using Raspberry Pi**. This work involved integrating planning, localization, sensor fusion, and control with a physical robotic platform, providing me with strong experience in translating research algorithms into practical autonomous robotic systems. My experimental work is available at <https://doi.org/10.5875/czmm9z84>.

My extensive work with **NSGA-II and multi-objective optimization** has provided me with a strong foundation for autonomous system development. Autonomous systems must continuously make decisions while balancing competing objectives such as **safety, efficiency, trajectory quality, obstacle avoidance, motion smoothness, and operational constraints**. I believe this experience is particularly relevant to intelligent vehicle systems and ADAS, where robust decision-making and system-level optimization are essential.

I am also currently working on **autonomous mobile robot systems with research objectives relevant to Level 4 (ADAS) automated driving concepts**. My work focuses on autonomous navigation and decision-making within a defined operational environment, where the robot is expected to perform its task without continuous human intervention. In this research, I am investigating my proposed improved NSGA-II approach for multi-objective path planning and decision-making while considering safety, obstacle avoidance, path efficiency, motion smoothness,

and environmental constraints. This work is helping me investigate system-level challenges that are closely related to higher levels of vehicle automation.

A significant part of my experience involves **sensor fusion and localization for autonomous mobile robots using ROS 2 and real-world experimental validation**. I have worked with multiple sensors, including IMU, wheel odometry, infrared sensors, drag wheel, and barometer, and developed adaptive Kalman filtering and multi-sensor fusion approaches for reliable pose localization. I have also implemented motion and PD-based control approaches to improve robot navigation and trajectory-tracking performance. My work emphasizes experimental validation on physical mobile robots rather than being limited to simulation.

Prior to joining academia, I worked as a **Robotics Software Engineer at WCB Robotics India Pvt. Ltd.**, where I developed algorithms for localization, SLAM, sensor fusion, control systems, robot kinematics, and coverage path planning for real-world façade-cleaning robotic systems. I worked on pose localization using adaptive Kalman filtering and multi-sensor fusion involving IR, IMU, odometry, drag wheel, and barometer data. This provided me with practical experience in taking robotic algorithms from development and simulation through system integration and experimental validation on physical robotic platforms.

My technical expertise includes **C, C++, Python, ROS 2, MATLAB, Webots, Gazebo, Git/GitHub**, and robotic system development. Through my research and industrial experience, I have developed a strong understanding of the complete autonomous-system pipeline, including **sensing, sensor fusion, state estimation, localization, planning, multi-objective optimization, decision-making, control, system integration, and real-time experimental validation**.

I am particularly interested in this position because it provides an opportunity to apply my autonomous robotics experience to **automotive system engineering and ADAS development**. While my primary experience has been in mobile robotics, the fundamental engineering challenges are closely related: integrating multiple sensors, handling uncertainty, estimating system state, making decisions in complex environments, generating safe and efficient trajectories, and validating autonomous behavior on physical systems.

I believe my combination of **Ph.D.-level research, industrial robotics experience, rigorous hands-on experience with autonomous mobile robots, ROS 2 development, sensor fusion, localization, improved NSGA-II-based optimization, autonomous navigation, and experimental validation** would allow me to contribute effectively to your system development team. I am particularly motivated to bring my experience in autonomous robotic systems into the automotive domain and further develop my expertise in ADAS and automated driving technologies.

I would welcome the opportunity to discuss how my research background, industrial experience, and technical expertise can contribute to your organization. Thank you for considering my application.

Sincerely,

Dr. Monex Sharma

Assistant Professor | Robotics Software Engineer

Ph.D., NIT Warangal

C/C++ | Python | ROS 2 | Sensor Fusion | Localization | Autonomous Navigation | Control Systems |
Multi-Objective Optimization